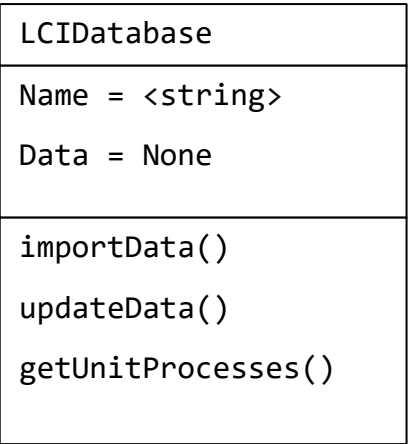
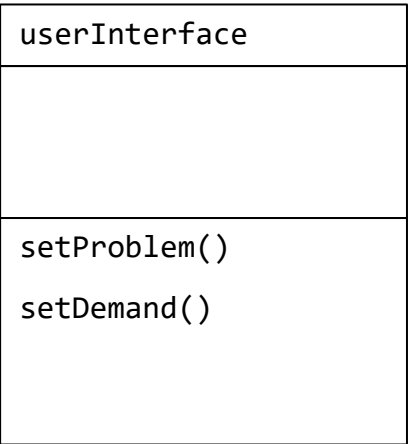


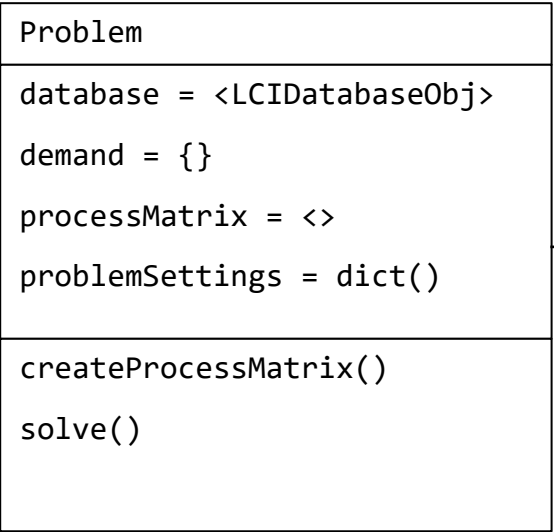


Abstract class with specific implementation based on the nature of how the data is provided.

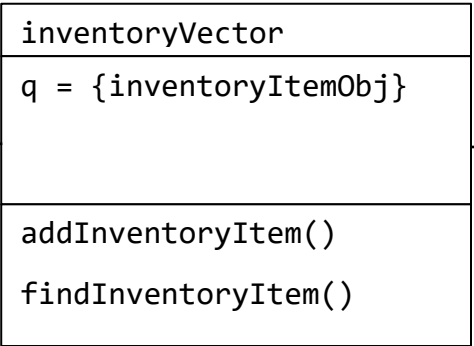
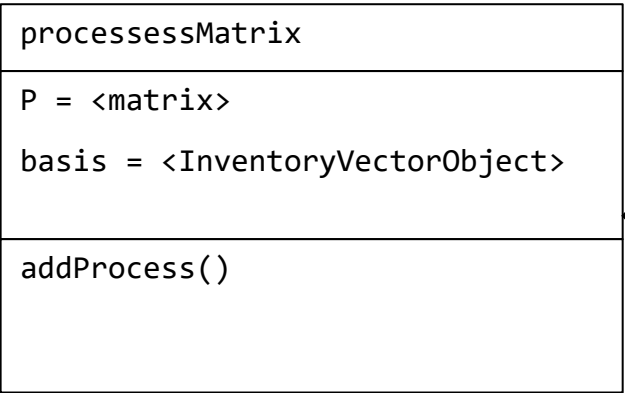
Does not store data---retrieves data from the data input.



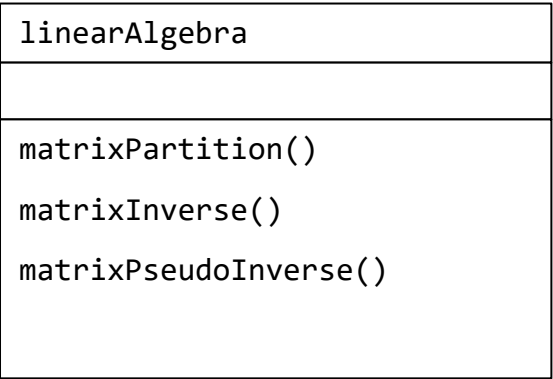
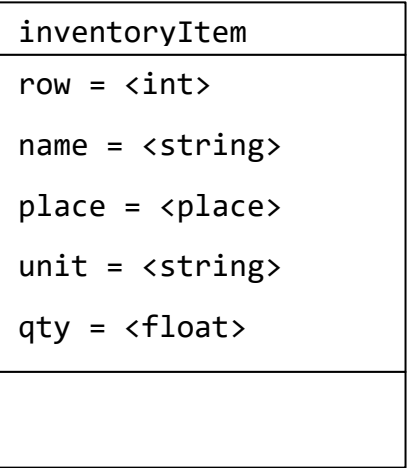
Abstract class with specific implementation based on the form of the interaction: e.g., console, GUI, Excel, etc.



creates



If inventory item name is unique, q could be a dictionary, making the search much faster.



creates

0..*

Pseudocode for methods

Problem

```
createProcessMatrix()
|   create empty processMatrix()
|   from Database getUnitProcessess()
|   for each unitProcess:
|       for each exchange():
|           if exchange not in q:
|               addInventoryItem()
|           addProcess(exchanges)
```

```
Solve()
|   f = getDemand()
|   A, B = partition processMatrix
|   if A is invertible:
|       s = inv(A) * f
|   else:
|       s = pinv(A) * f
|   g = B * s
|   update inventoryVector
```

processessMatrix

```
addProcess(exchanges)
|   rows = find exchanges in inventoryVector
|   add new zero column
|   put exchangeQtys in rows
```